

INNOCENT KELEBOGILE MARITI

AI AUTOMATION ENGINEER

Johannesburg, Gauteng, South Africa | innocentmariti@gmail.com
github.com/JuiceMatute356 | linkedin.com/in/innocent-mariti-27a4b552 | maritico.co.za

PROFESSIONAL SUMMARY

Automation engineer who builds and runs LLM-powered workflow systems in production, on infrastructure I administer myself. Currently 12 workflows live across 2,172 nodes on a self-hosted stack, carrying 2,581 executions in the last seven days with one hard failure, on a server with 128 days of continuous uptime. The work is customer-facing and money-facing, so it is built to fail closed: every release goes out behind automated guard tests, negative controls and a rollback point. Twelve years selling and structuring vehicle finance deals is where the problems come from, and it means I can run discovery with a non-technical owner and then go and build the thing. No computer science degree. The systems are live and the evidence is public.

EVIDENCE IN PRODUCTION

- **Reliability:** 2,581 workflow executions in seven days with one hard failure, across 12 active workflows, 34 webhooks and 33 credentials. I quote the raw counts rather than an uptime percentage, because a completed execution is not proof of a correct outcome. Verified 2026-08-20.
- **Scale of build:** 54 workflows authored, 2,172 nodes in total. The flagship conversational system is a single 489-node workflow.
- **Cost engineering:** cut LLM full-price input tokens from 12,874 to 2,801 per conversation turn, a 78% reduction, by restructuring prompt-cache breakpoints around what was genuinely volatile per turn instead of what a comment claimed was.
- **Latency:** reduced perceived reply time from 22.8s to 14.7s, a 35% cut, using a per-node execution profiler I wrote to find where the time actually went.
- **Correctness under money risk:** found and fixed a finance calculation quoting 30 to 33% light across 23 vehicle derivatives, then built a guardrail that refuses to release a figure it cannot tie to a specific model. Verified 28 of 28 cases against live production data plus 17 of 17 adversarial cases.
- **Testing discipline:** 187 server-side assertions gate a CRM release, and a guard is not trusted until it has been proven able to go red. Three guards were found decorative, passing regardless of the state they claimed to check.
- **Uptime and recovery:** 128 days continuous uptime, services self-healing under systemd, a dead man switch on a 5-minute cron, and a snapshot plus rollback point taken before every production change.

AUTOMATION AND AI ENGINEERING

Automation Engineer, Independent | MaritiCo

2025 - Present | Johannesburg, remote | Motor retail, healthcare and professional services, client names under NDA

- Design, build and operate production automation on self-hosted n8n, Python and JavaScript, owning the work end to end from discovery through to running it in production.
- Build conversational systems on the WhatsApp Cloud API and Telegram Bot API: multi-turn context held in Redis, intent routing, human approval gates before a message reaches a customer, and delivery proven against provider message IDs rather than trusting a green execution.
- Integrate LLM APIs (Anthropic Claude, OpenAI, DeepSeek) into live workflows for classification, structured extraction, drafting and document handling, with prompt-cache engineering to control cost per turn.
- Integrate REST APIs across Meta Graph, Google Workspace via OAuth2 (Sheets, Docs, Drive, Calendar, Gmail), Twilio, SMTP and IMAP, and dealer management systems.
- Administer the full stack: Ubuntu VPS, Docker and Docker Compose, nginx, Redis, SQLite, Cloudflare Tunnel and Zero Trust, systemd, cron, backups and incident recovery.
- Own the security side: secrets moved out of workflow bodies into environment-scoped credentials, credential rotation, HTTPS and HSTS enforcement, and an authentication gap on a public webhook found and closed by testing it from outside the network.
- Write the monitoring: health checks, drift detection between duplicated logic, token-expiry watchers and scheduled digests that alert before a client notices a failure.
- Run discovery with non-technical owners and managers, translate the business problem into a workflow design, and document it so another engineer can pick it up.

SELECTED PROJECTS

Conversational Sales Assistant (WhatsApp, LLM, 489 nodes, live)

- Answers inbound vehicle enquiries in natural language, holds conversation context across a thread, books test drives and service, and runs a finance application through to completion with document capture.
- Twelve independent feature gates let any lane be switched off without a redeploy. A pricing guardrail holds any monthly figure it cannot tie to a named model, so a wrong number never reaches a customer.

Finance and Deal CRM (TypeScript, React, REST)

- Role-based staff CRM for structuring and tracking vehicle finance deals. Fail-closed by design: no route can fabricate a credit decision, outbound is paused by default, and 187 server-side assertions cover the guards, field projection, record ownership and the money calculations.

Grounded Retrieval (RAG) with a Measured Refusal Gate

- Document retrieval over a policy corpus using local ONNX embeddings and a SQLite vector store, with no external vector database and no embedding API.
- Built an eval harness measuring recall and refusal accuracy across 38 labelled questions. Retrieved the right document for 19 of 20 answerable questions and correctly refused 10 of 10 off-topic ones. Published the honest 81.6% overall rather than the clean sweep an easier first eval set produced.

Field Inspection to Approval Pipeline

- Phone-based inspector captures items and photographs, multiple approvers action it by email, and an approved job produces a PDF work order for the workshop.

SALES EXPERIENCE

New Vehicle Sales Executive | Motor Retail Group

August 2026 - Present | West Rand, Gauteng

- New vehicle sales across a fast-moving brand range, working inbound digital leads through to delivery.

Sales Executive | Hyundai Motor Company

November 2025 - July 2026 | Midrand, Gauteng

- Averaged 8 units a month against a 6-unit target (133%), roughly R110,000 gross profit a month, ranked 3rd of 9 on the floor.
- Converted 90% of deals to finance, structured through Signio and Seriti under NCA and F and I requirements.
- Built and ran the internal finance-application and stock track-and-trace tools used by the used-car team.

Earlier Sales Roles

June 2014 - October 2025 | Career break June 2022 to February 2024

- Sales Executive, Williams Hunt Fourways (Mar 2024 to Oct 2025). Structured compliant finance deals and assessed credit risk across the full purchase cycle.
- Sales Executive, iCars Technologies / Digi-Cars (Sep 2020 to May 2022). End-to-end sales on a digital-first platform, working leads from AutoTrader and Cars.co.za.
- Sales Executive, Mercedes-Benz (Aug 2017 to Jul 2020). Premium customer base, monthly volume and CSI targets under OEM brand standards.
- Sales Executive, IC Auto Nissan (Jun 2014 to Aug 2017). Full sales process from first enquiry to delivery, building repeat and referral business.

TECHNICAL SKILLS

Automation and orchestration: n8n (self-hosted), webhook and event-driven design, retry and idempotency, queue and state management, cron scheduling

AI and LLM: Anthropic Claude, OpenAI, DeepSeek, system prompt design, prompt caching and cost control, structured output extraction, output guardrails and validation, conversational state

Retrieval and RAG: embeddings, vector search, chunking strategies, retrieval evaluation and threshold tuning, grounded answering with citations, refusal gating

Languages: Python, JavaScript, TypeScript, SQL, Bash

APIs and integration: REST, webhooks, OAuth2, Meta Graph and WhatsApp Cloud API, Telegram Bot API, Google Workspace, Twilio, SMTP and IMAP

Infrastructure: Linux (Ubuntu) VPS administration, Docker and Docker Compose, nginx, Cloudflare Tunnel and Zero Trust, systemd, Redis, SQLite, Git, backup and restore

Reliability and security: monitoring and alerting, health checks, drift detection, automated guard tests and negative controls, incident recovery, secrets management and credential rotation, HTTPS and HSTS enforcement

Commercial: stakeholder discovery with non-technical users, process mapping, requirements translation, technical documentation and handover, NCA and POPIA aware record handling

EDUCATION

- Matric (Grade 12), Seqhobong Secondary School, 2009
- NCA Accreditation (National Credit Act, vehicle finance)
- Automation and AI engineering, 2025 to present. Learned by building and operating production systems. Work is public at github.com/JuiceMatute356

REFERENCES

Available on request. Referee contact details withheld in line with POPIA.